Lec-16

19/04/26 9:45 AM

Substring

coding →

c co cod codi codin coding

o od odi odin odin

d di din dins

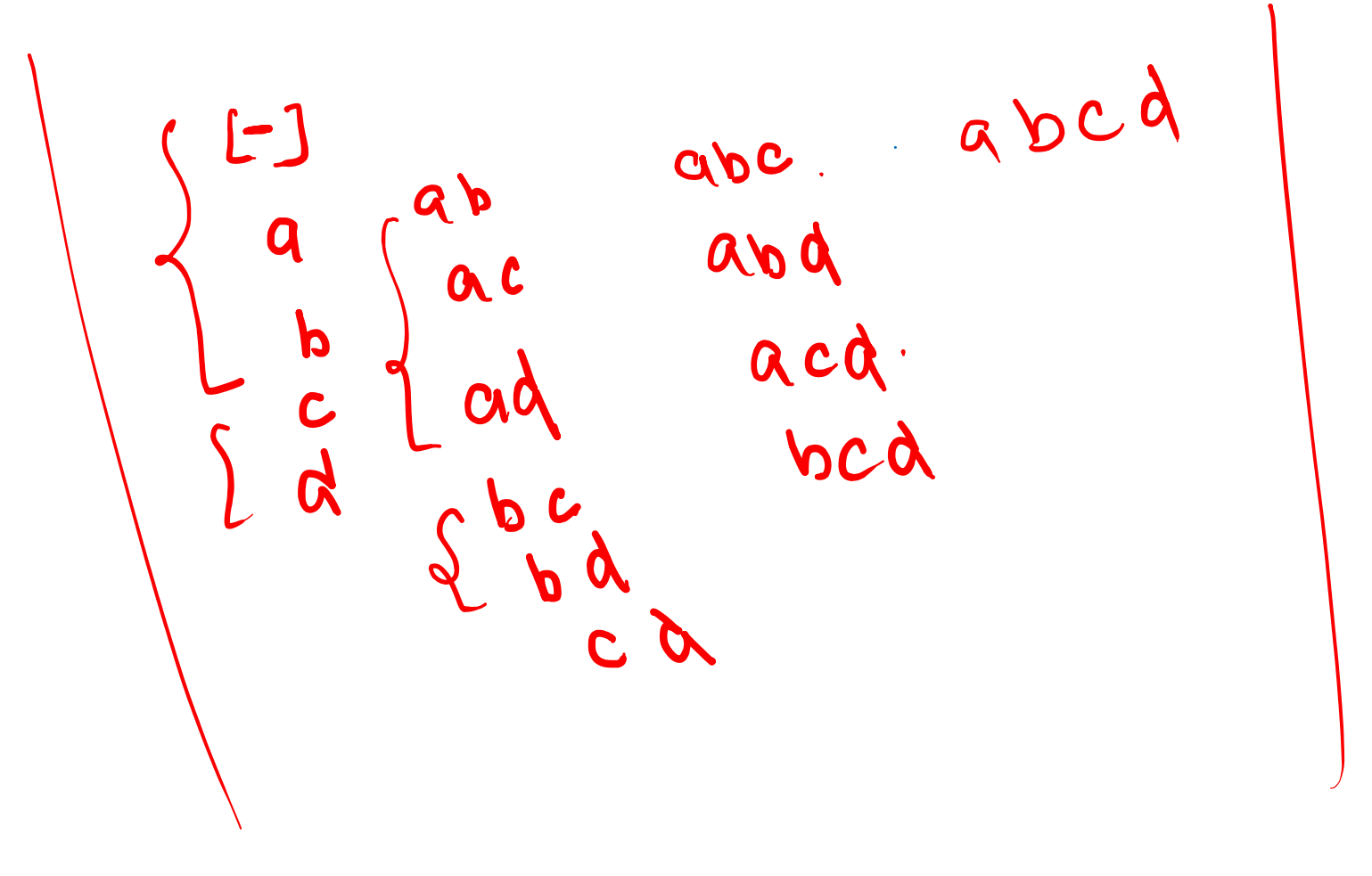
i in ins

n n)

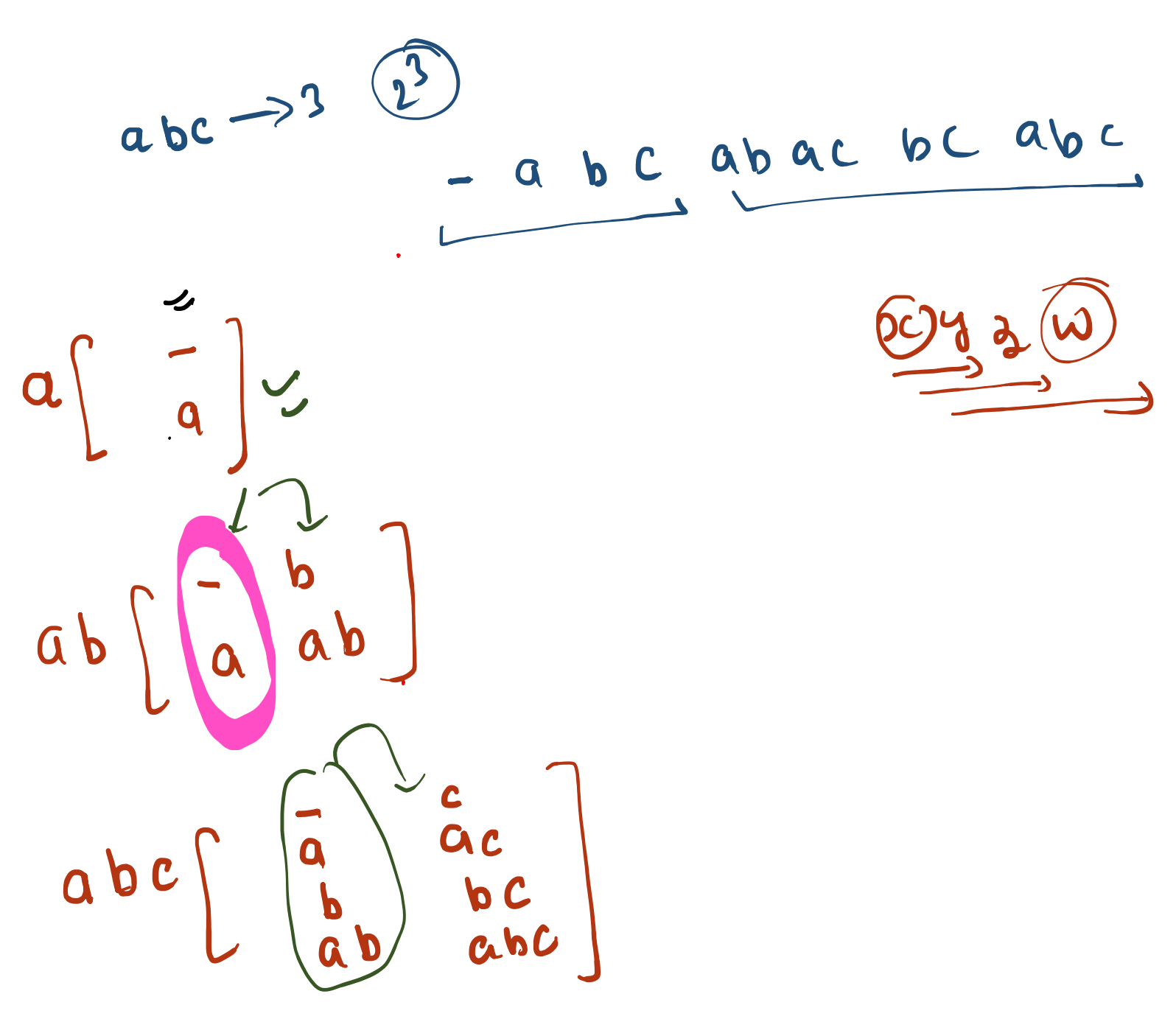
Subsring → Subsequence

Subseqence

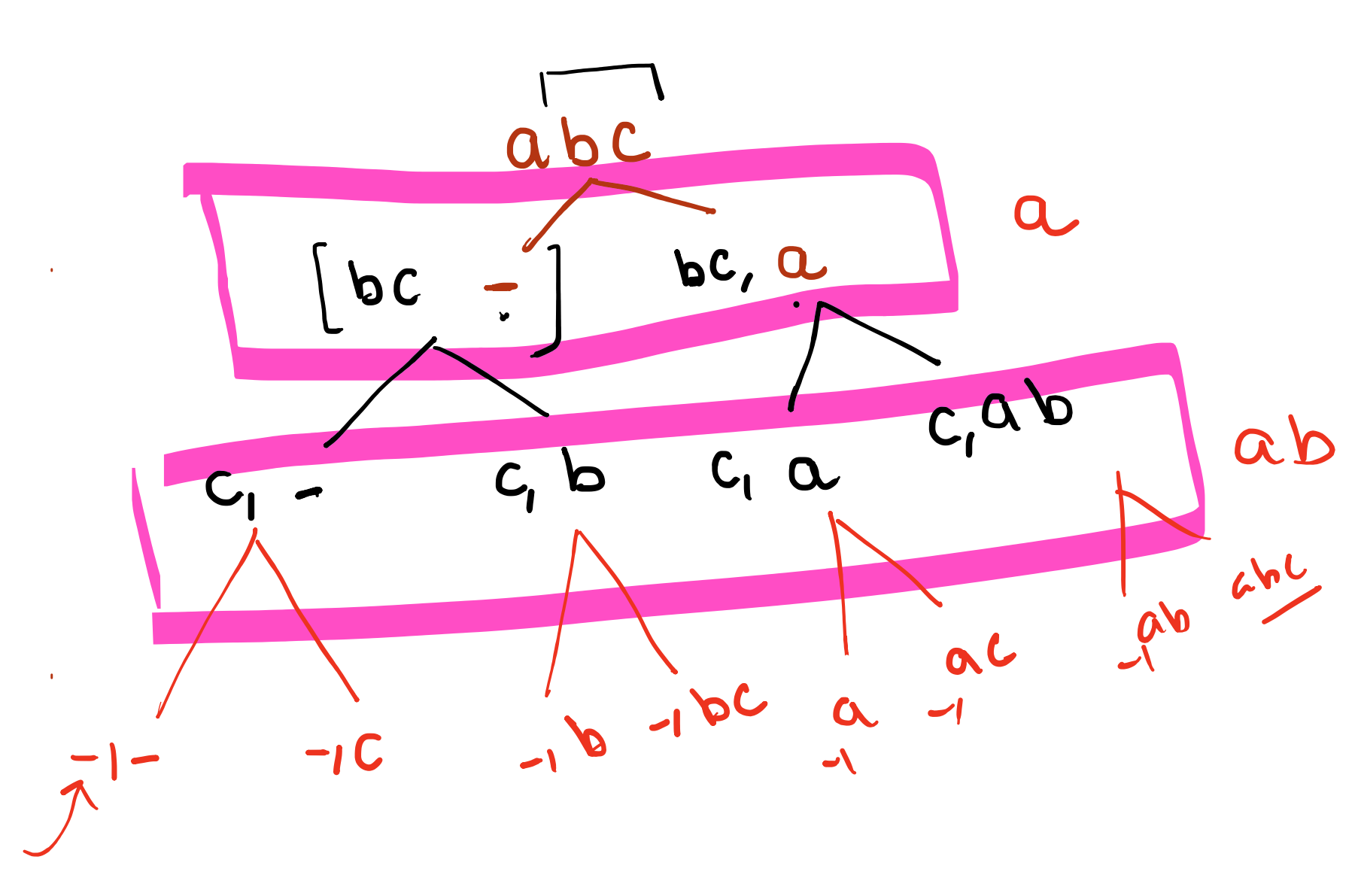
abcd



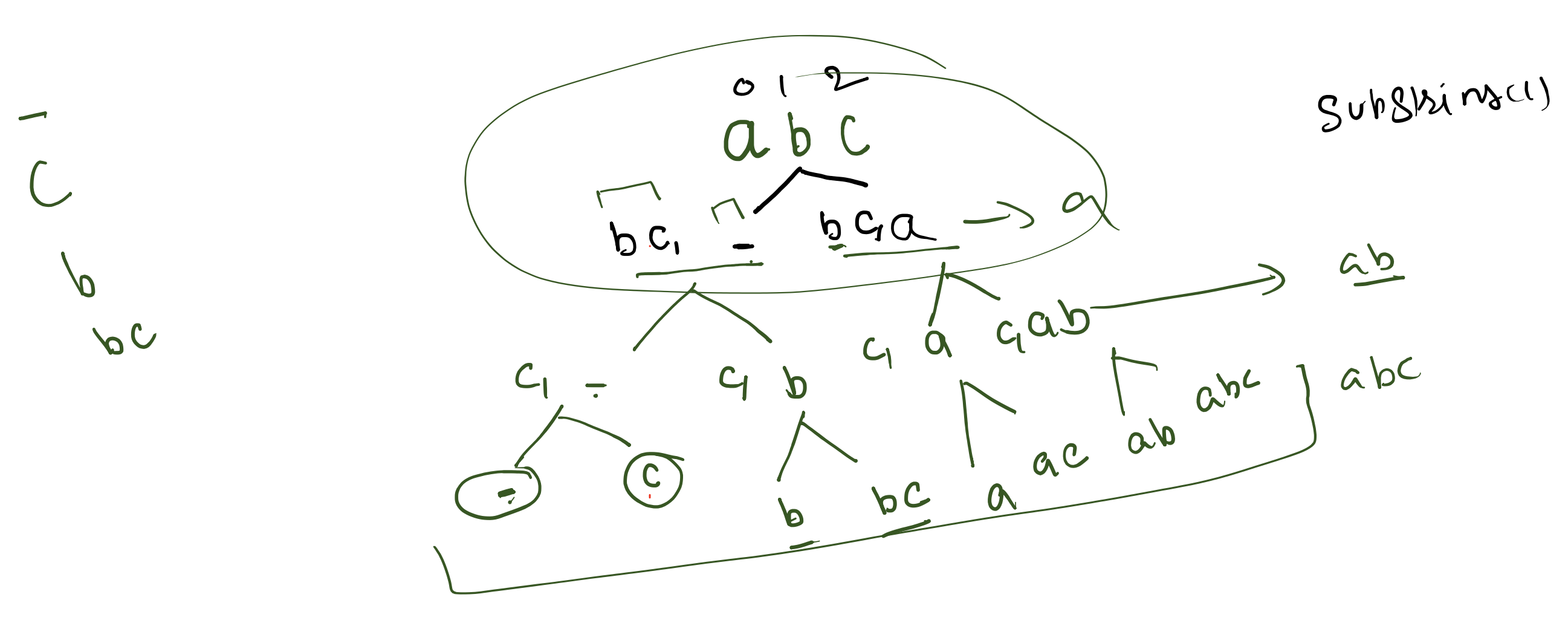
[-]
a
b
c
d
ab
ac
ad
bc
bd
cd
abc
abd
acd
bcd
abcd



abc → 3 (2^3)
- a b c ab ac bc abc
(x)y 2 (w)
a
ab
abc
-
a
b
ab
c
ac
bc
abc

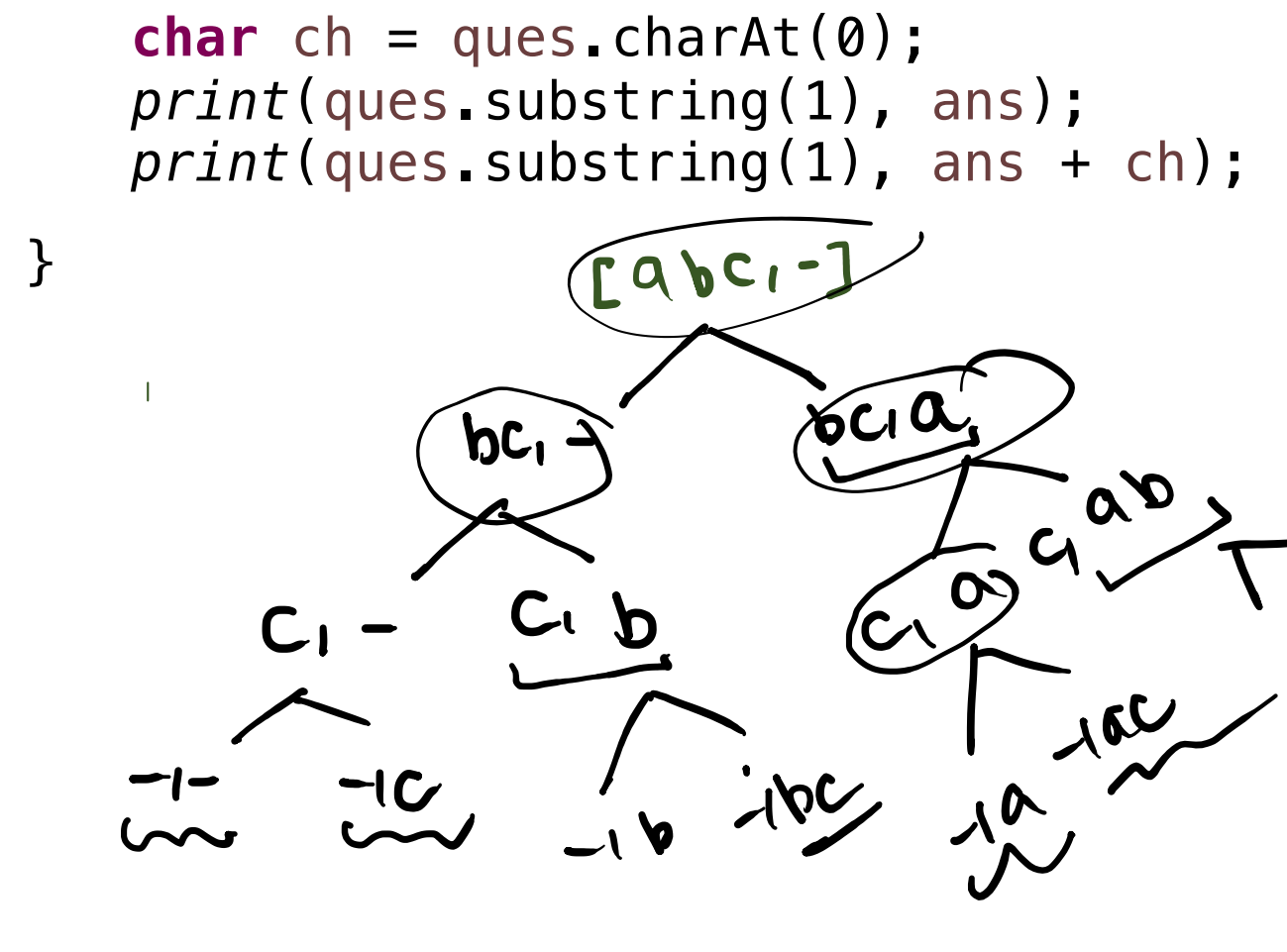


abc
[bc -]
bc, a
a
c, -
c, b
c, a
c, ab
ab
-, -
-, c
-, b
-, bc
a
ac
-, ab
abc

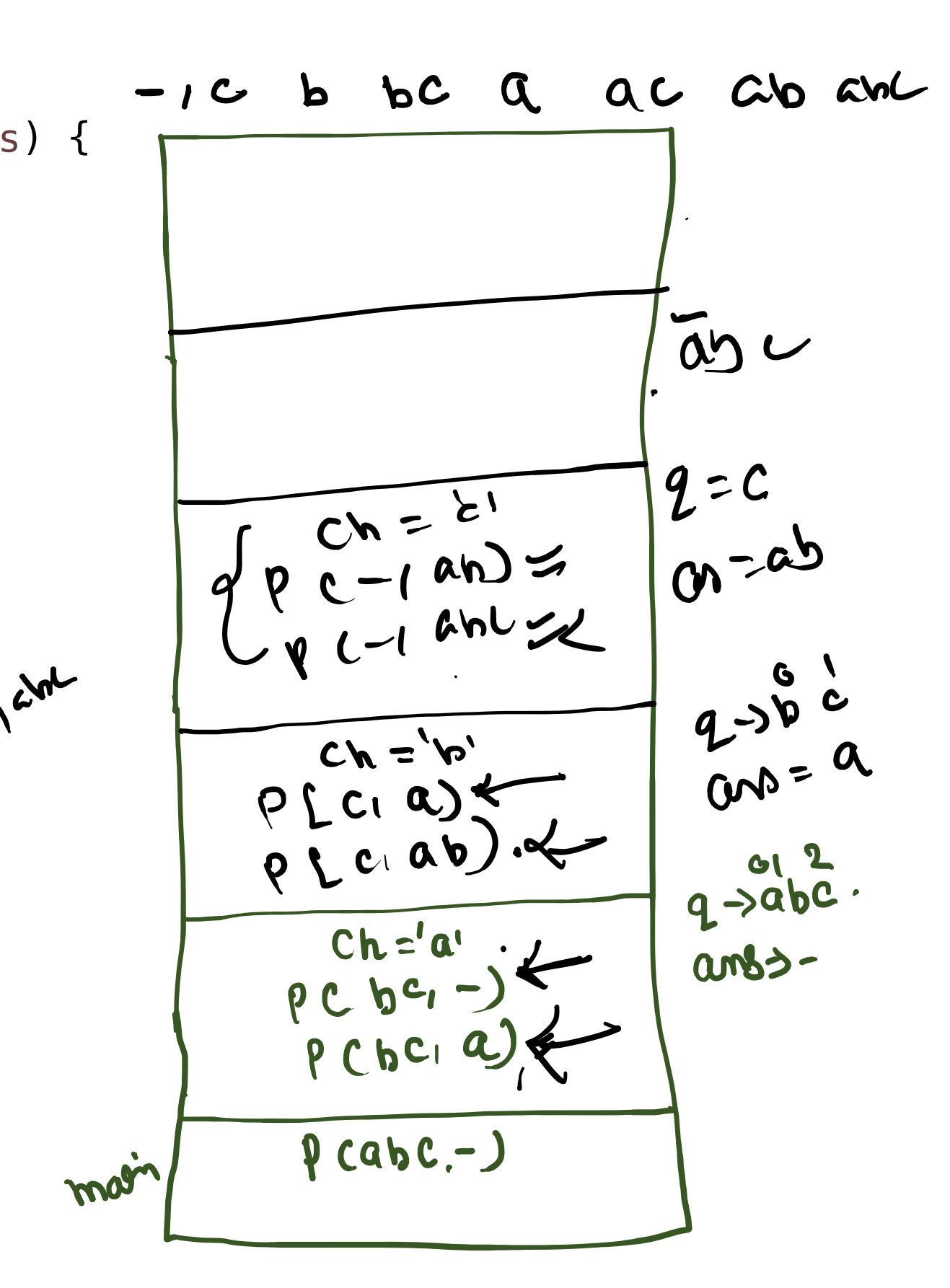


-
c
b
bc
0 1 2
abc
bc, -
bc, a
a
c, -
c, b
c, a
c, ab
ab
-
c
b
bc
a
ac
ab
abc
abc
Substring(1)

```
public static void print(String ques, String ans) {
		if (ques.length() == 0) {
			System.out.print(ans + " ");
			return;
		}
		char ch = ques.charAt(0);
		print(ques.substring(1), ans);
		print(ques.substring(1), ans + ch);
	}
```

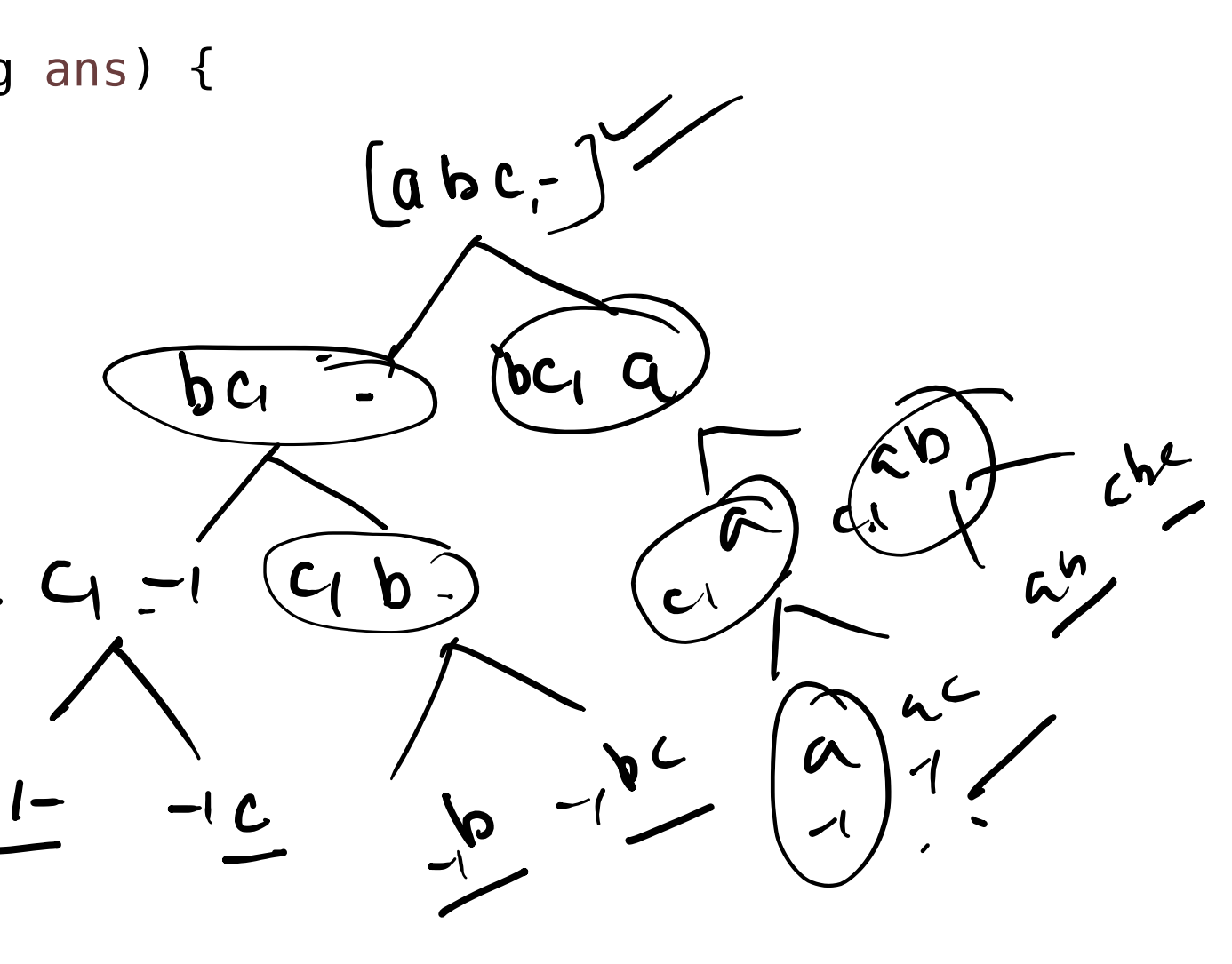


[abc, -]
bc, -
bc, a
c, -
c, b
c, a
c, ab
-, -
-, c
-, b
-, bc
-, a
-, ac
-, ab
-, abc



-, c b bc a ac ab abc
abc
q = c
an = ab
ch = 'c'
p(-, ab)
p(-, abc)
q → bc
ans = a
ch = 'b'
p(c, a)
p(c, ab)
q → abc
ans → -
ch = 'a'
p(bc, -)
p(bc, a)
p(abc, -)
main

```
public static void print(String ques, String ans) {
		if (ques.length() == 0) {
			System.out.print(ans + " ");
			return;
		}
		char ch = ques.charAt(0);
		print(ques.substring(1), ans);
		print(ques.substring(1), ans + ch);
	}
```



[abc, -]
bc, -
bc, a
c, -
c, b
c, a
c, ab
-, -
-, c
-, b
-, bc
a
ac
ab
abc

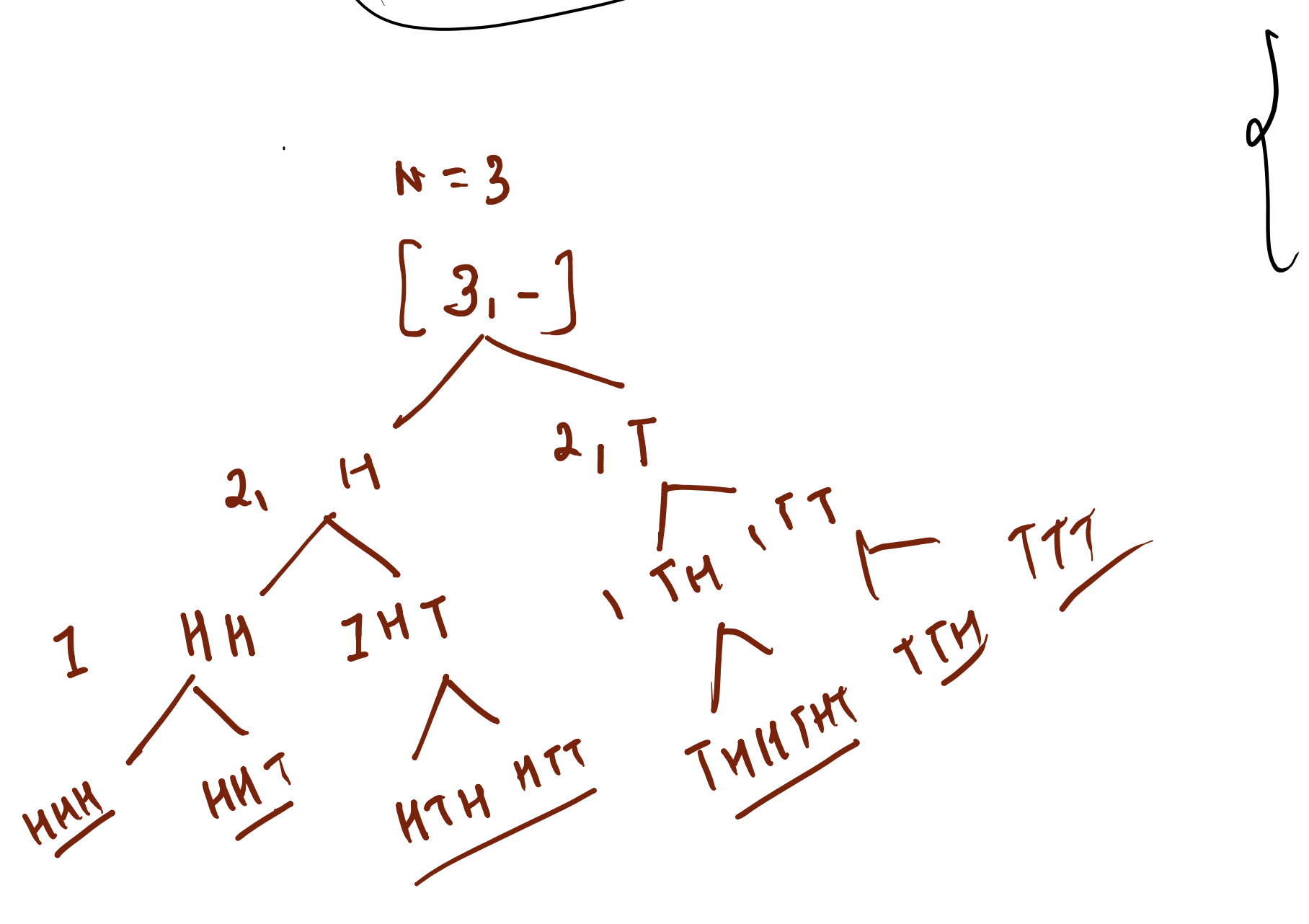
- c b bc a ac

an abc

N → coins

N = 3

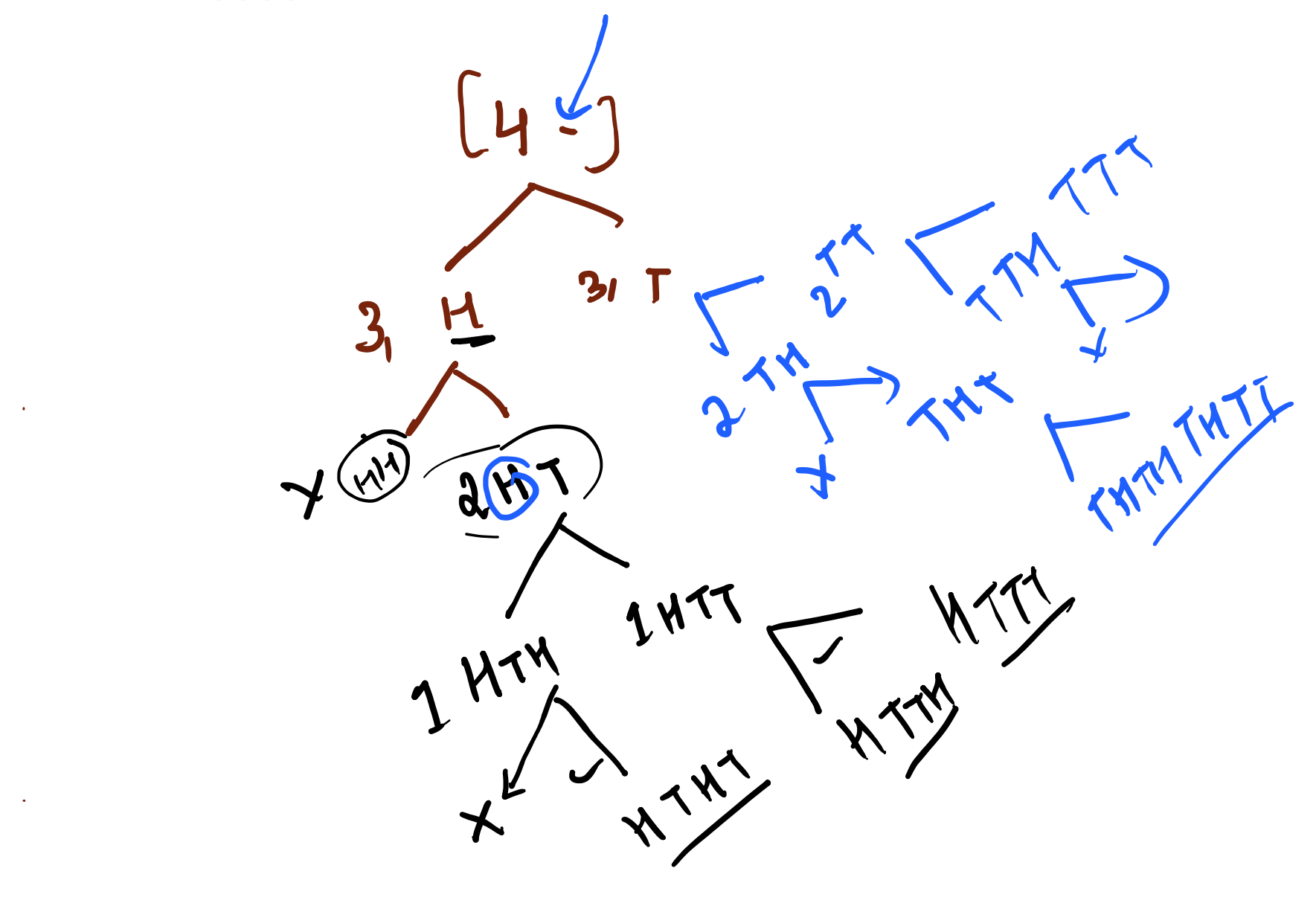
HHH
HHT
HTH
HTT
THH
THT
TTH
TTT



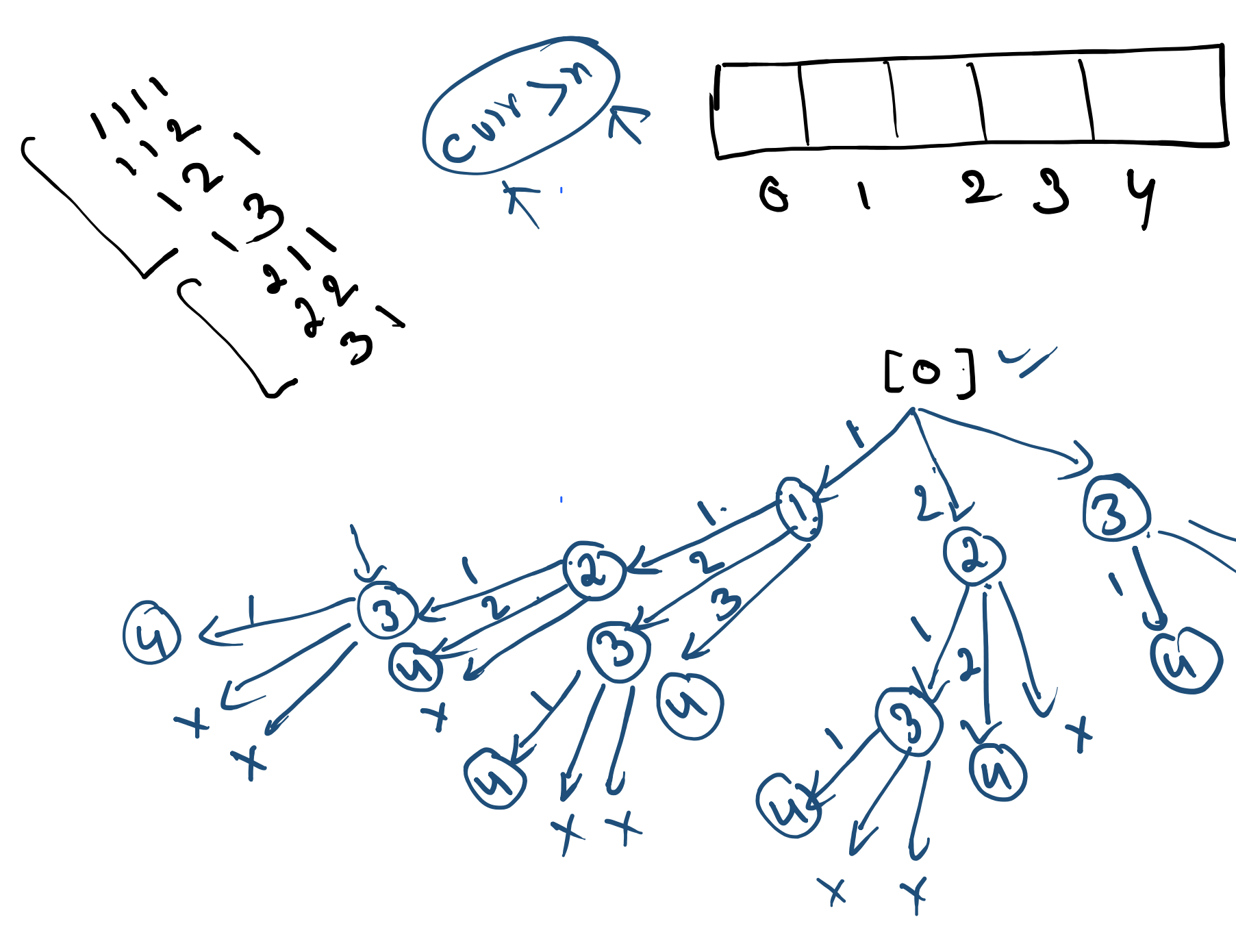
N = 3
[3, -]
2, H
2, T
1 HH
1 HT
1 TH
1 TT
HHH
HHT
HTH
HTT
THH THT
TTH
TTT

HHHH
HHHT
HHTH
HHTT
HTHH
HTHT
HTTH
HTTT
THHH
THHT
THTH
THTT
TTHH
TTHT
TTTH
TTTT

HHH
HHT
HTH
HTT
THH
THT
TTH
TTT
HHH
HHT
HTH
HTT
THH
THT
TTH
TTT



[4, -]
3, H
3, T
HH
2HT
2 TH
2 TT
THT
TTH
THTH THT
1 HTH
1 HTT
HTHT
HTTH
HTTT



1111
112
121
13
211
22
31
curr > n
0 1 2 3 4
[0]

→ Size N = 4

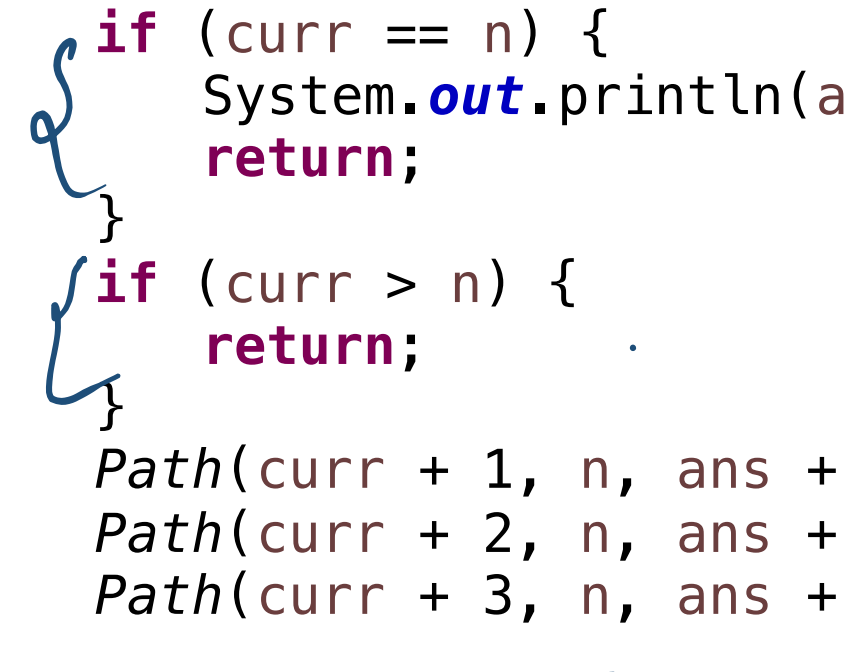
dice → [1, 2, 3, 4, 5, 6]

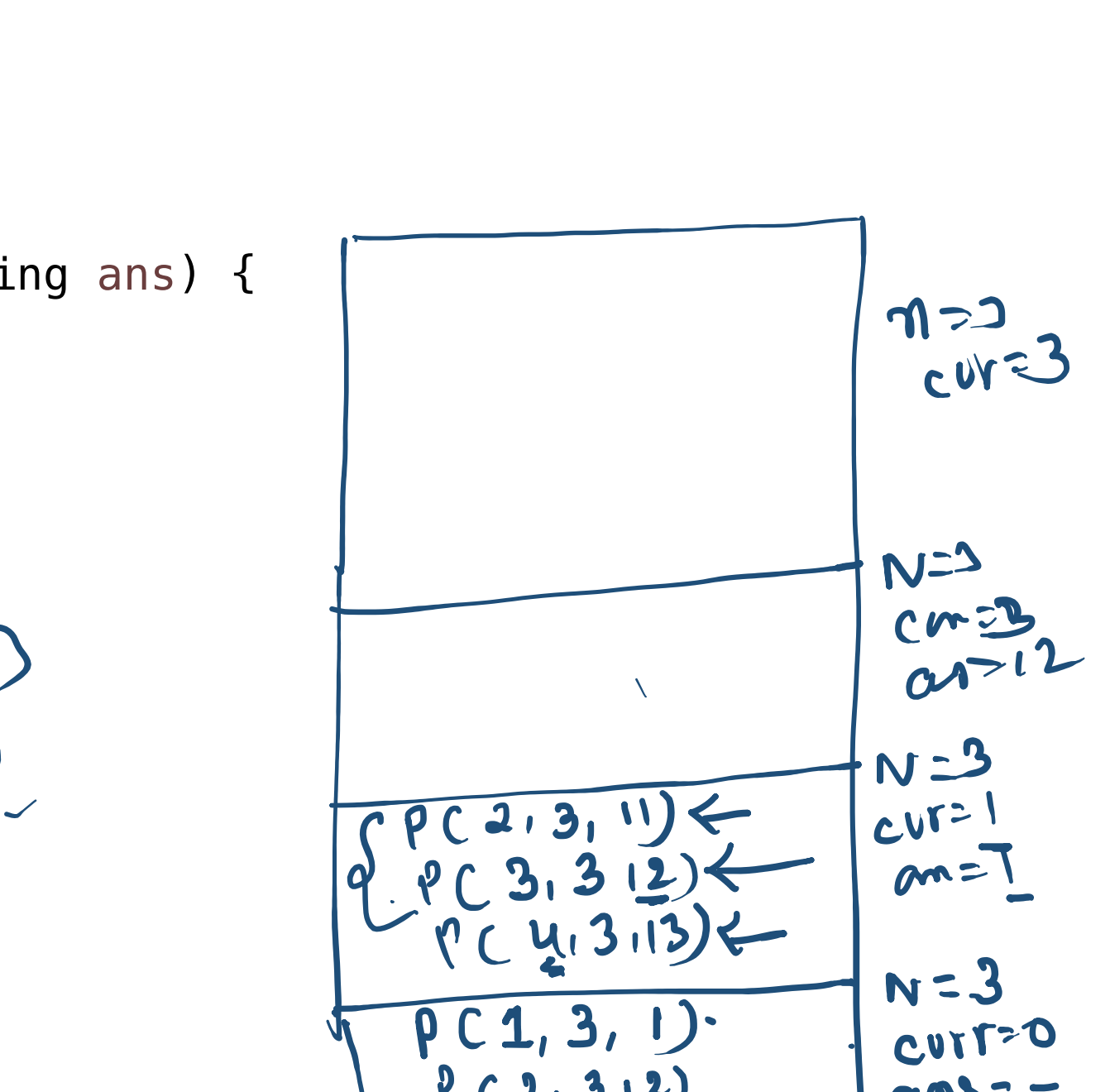
→ 3 [1, 2, 3]

1111
112
121
13
211
22
31

```
public static void Path(int curr, int n, String ans) {
		// TODO Auto-generated method stub
		if (curr == n) {
			System.out.println(ans);
			return;
		}
		if (curr > n) {
			return;
		}
		Path(curr + 1, n, ans + 1);
		Path(curr + 2, n, ans + 2);
		Path(curr + 3, n, ans + 3);
	}
```

111
12
21

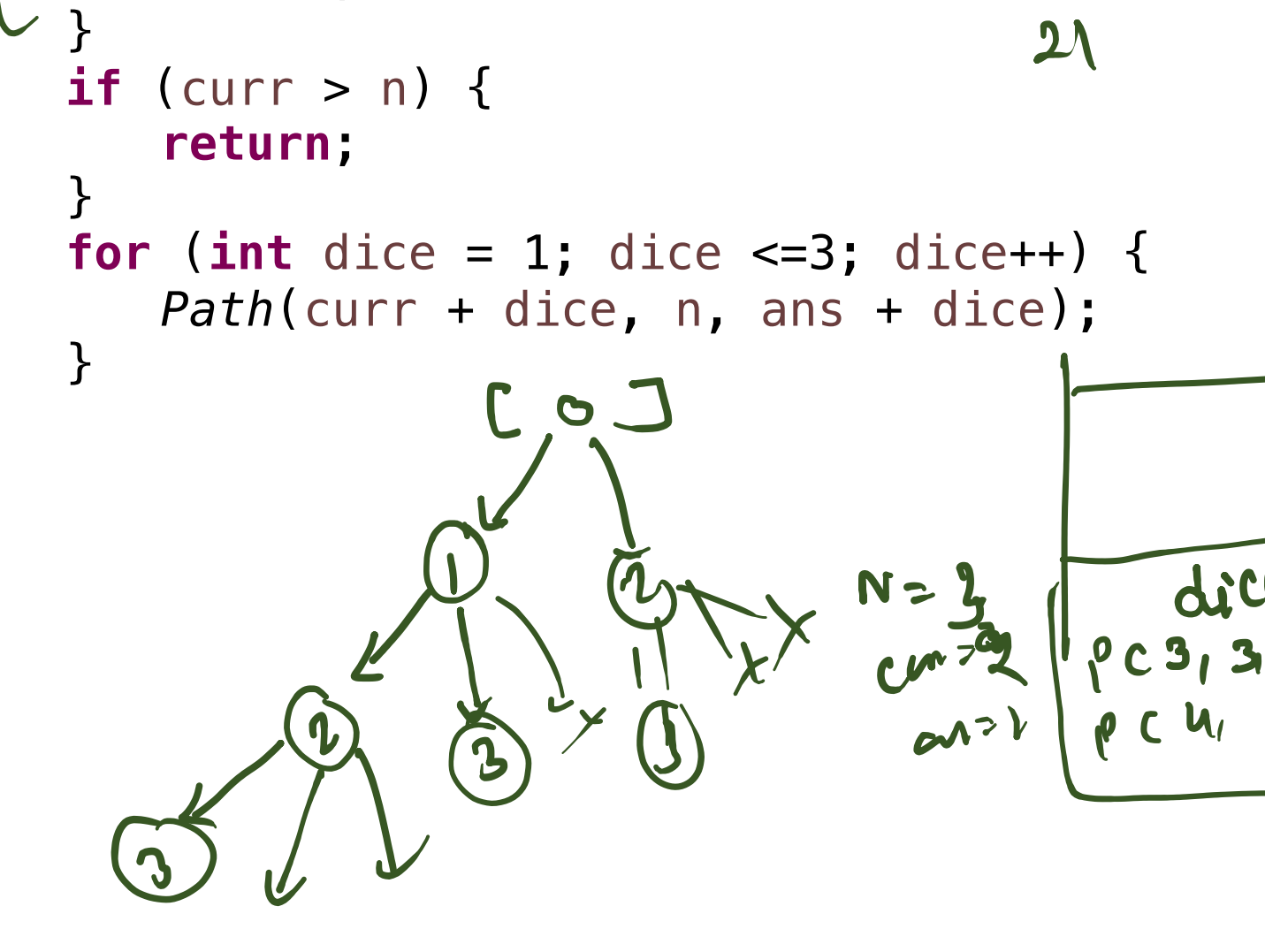




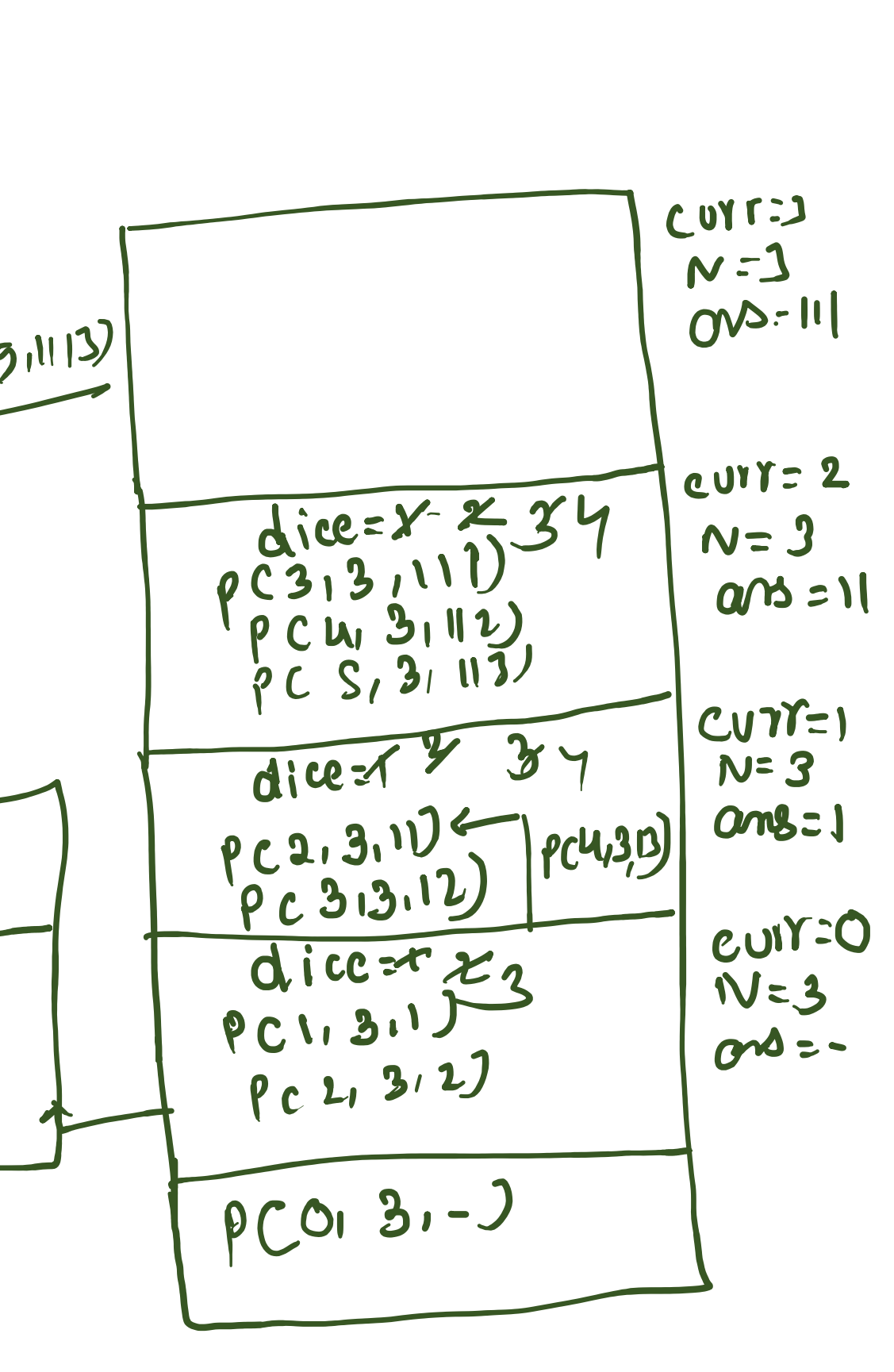
n = 3
curr = 3
N = 3
curr = 3
ans = 12
N = 3
curr = 1
ans = 1
N = 3
curr = 0
ans = -
P(2, 3, 1)
P(3, 3, 12)
P(4, 3, 13)
P(1, 3, 1)
P(2, 3, 2)
P(3, 3, 3)
P(0, 3, -)
main

```
public static void Path(int curr, int n, String ans) {
		// TODO Auto-generated method stub
		if (curr == n) {
			System.out.println(ans);
			return;
		}
		if (curr > n) {
			return;
		}
		for (int dice = 1; dice <=3; dice++) {
			Path(curr + dice, n, ans + dice);
		}
	}
```

111
12
21



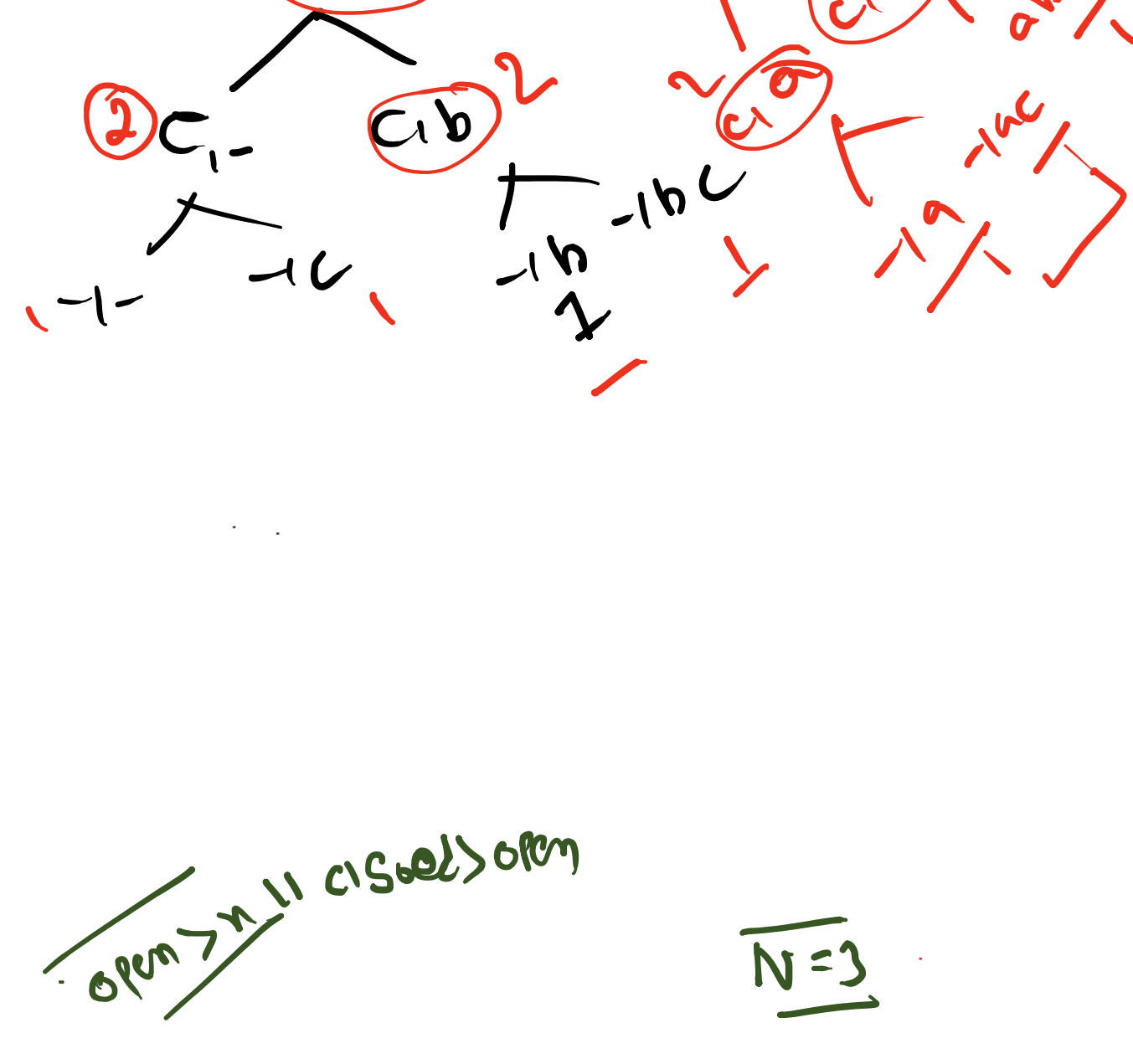
[0]
N = 3
curr = 2
ans = 1
dice = 1 2
P(3, 3, 21)
P(4,



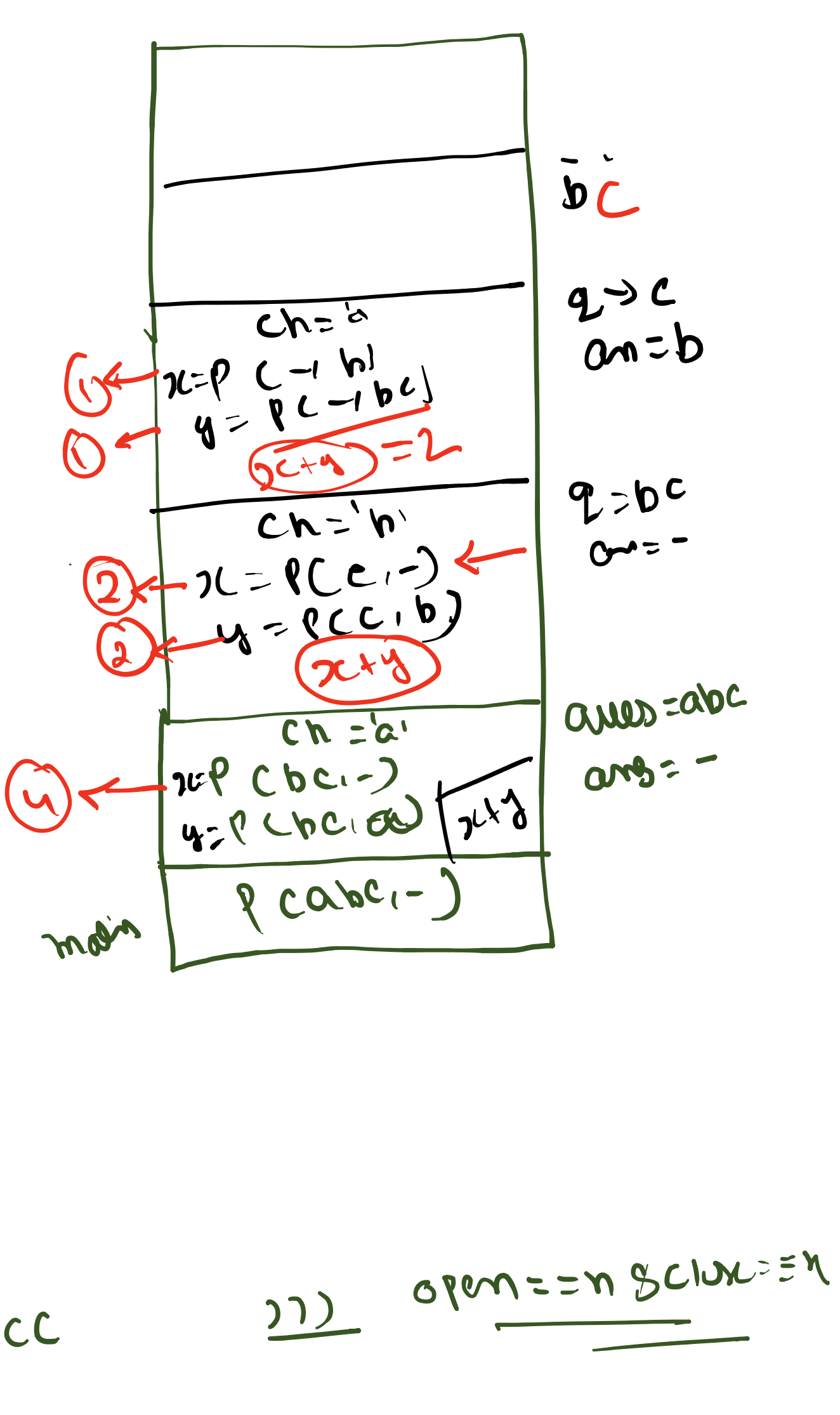
[2, 4, 7, 9, 11, 13]
curr = 3
N = 3
ans = 111
dice = 1 2 3 4
P(3, 3, 111)
P(4, 3, 112)
P(5, 3, 113)
curr = 2
N = 3
ans = 11
dice = 1 2 3 4
P(2, 3, 11)
P(3, 3, 12)
P(4, 3, 13)
curr = 1
N = 3
ans = 1
dice = 1 2 3
P(1, 3, 1)
P(2, 3, 2)
curr = 0
N = 3
ans = -
P(0, 3, -)

- c b bc

```
public static int print(String ques, String ans) {
		if (ques.length() == 0) {
			System.out.print(ans + " ");
			return 1;
		}
		char ch = ques.charAt(0);
		int x = print(ques.substring(1), ans);
		int y = print(ques.substring(1), ans + ch);
		return x + y;
	}
```



abc -
y = 4
4
bc -
bc, a
2
c, -
c, b
2
c, ab
c, a
2
-, -
-, c
-, b
-, bc
1
-, a
-, ac
ab
abc



bc
q → c
an = b
ch = 'c'
x = P(-, b)
y = P(-, bc)
x + y = 2
q = bc
an = -
ch = 'b'
x = P(c, -)
y = P(c, b)
x + y
ques = abc
ans = -
ch = 'a'
x = P(bc, -)
y = P(bc, a)
x + y
P(abc, -)
main

open > n || close > open

N = 3

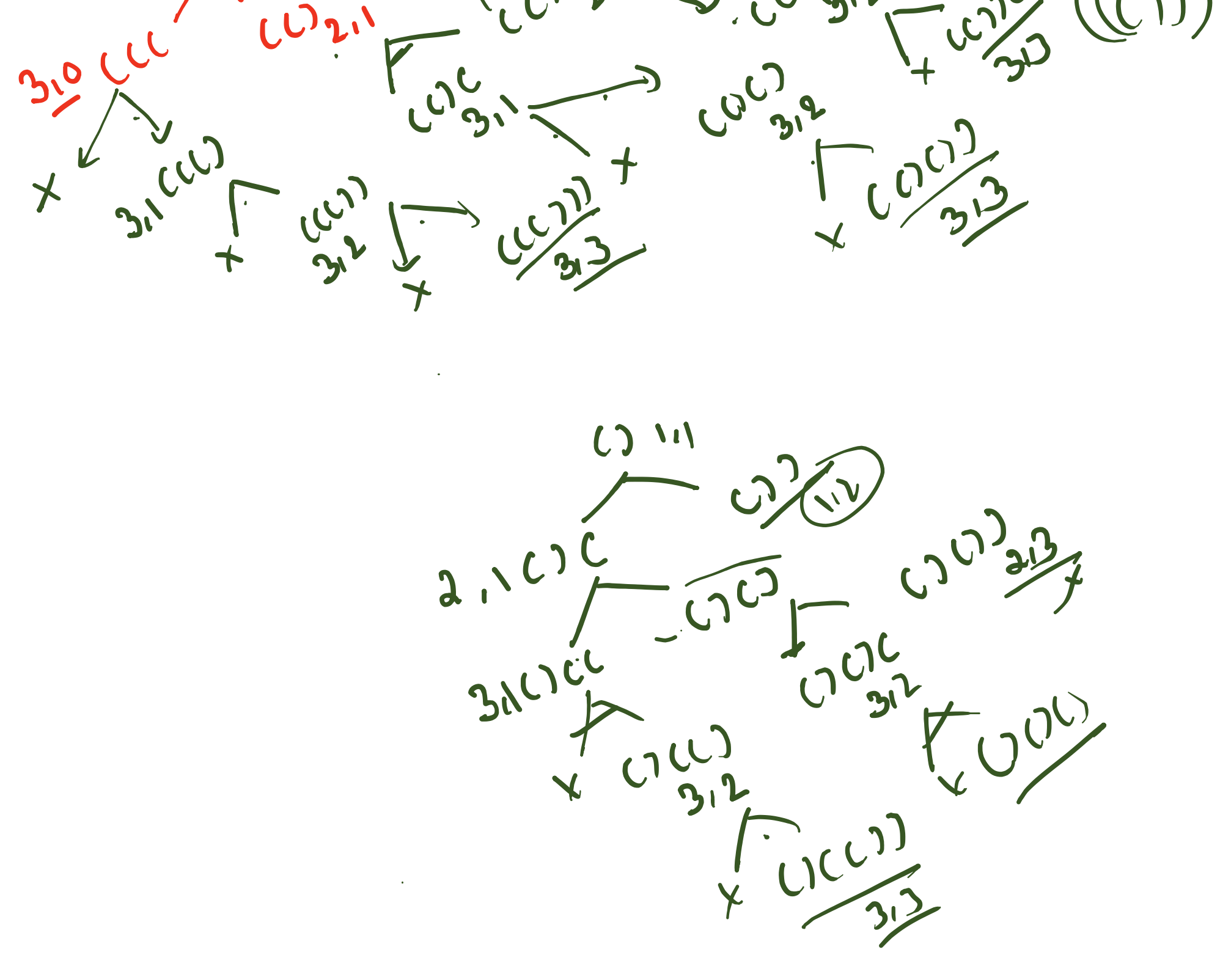
((()))
(()())
(())()
()(())
()()()

CCC

)))

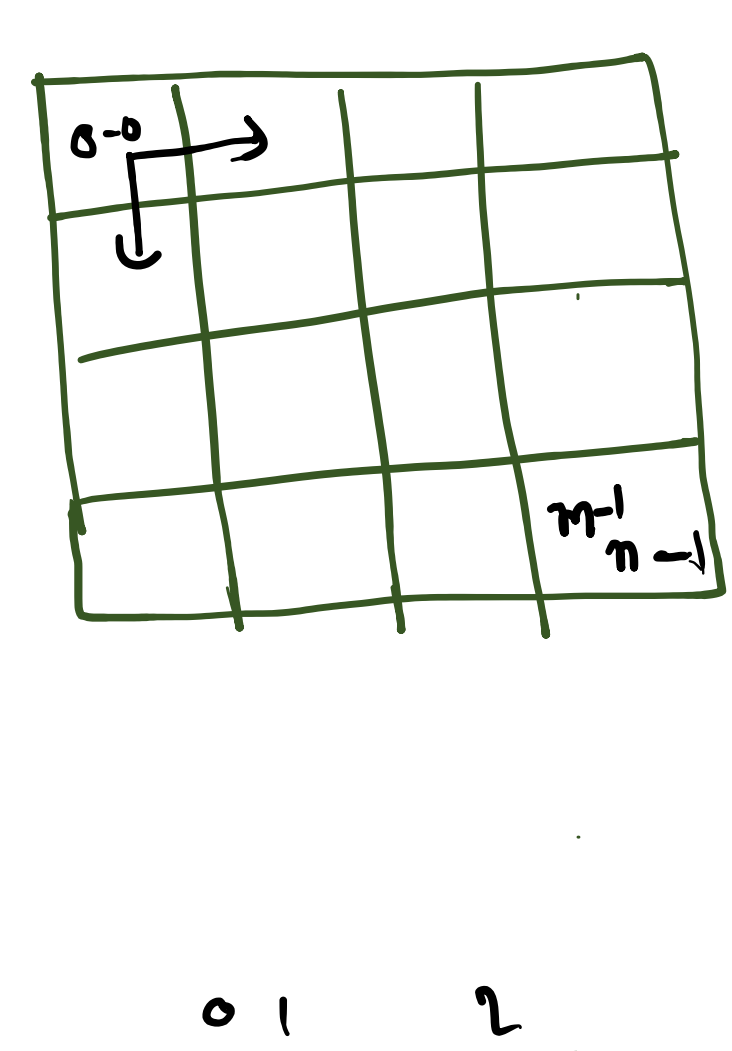
open == n & close == n

()()()
(()())
((())
()(())
((()))



[0, 0]
1, 0 (
0, 1)
2, 0 ((
()
1, 1
3, 0 (((
(()
2, 1
3, 1 ((()
(()(
3, 1
(())
2, 2
((())
3, 2
(()()
3, 2
(())(
3, 2
((()))
3, 3
(()())
3, 3
(())()
3, 3

() 1, 1
2, 1 ()(
()) 1, 2
3, 1 ()((
()()
2, 2
()())
2, 3
()(()
3, 2
()()(
3, 2
()(())
3, 3
()()()

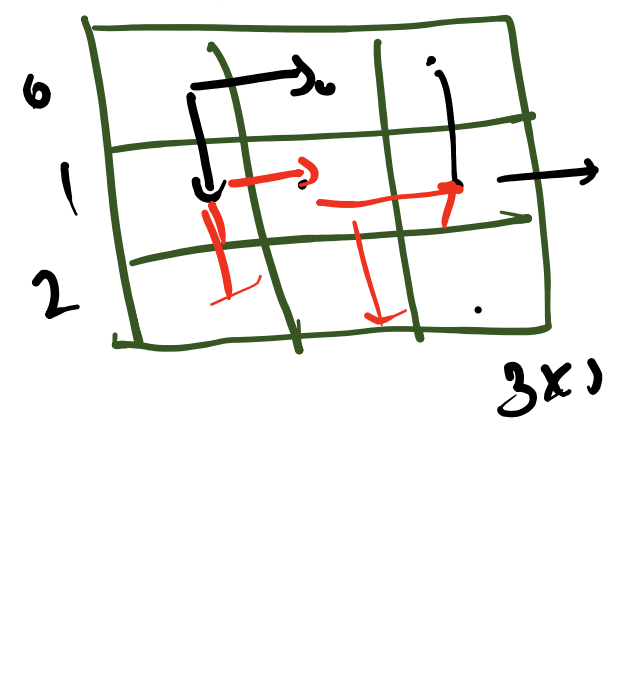


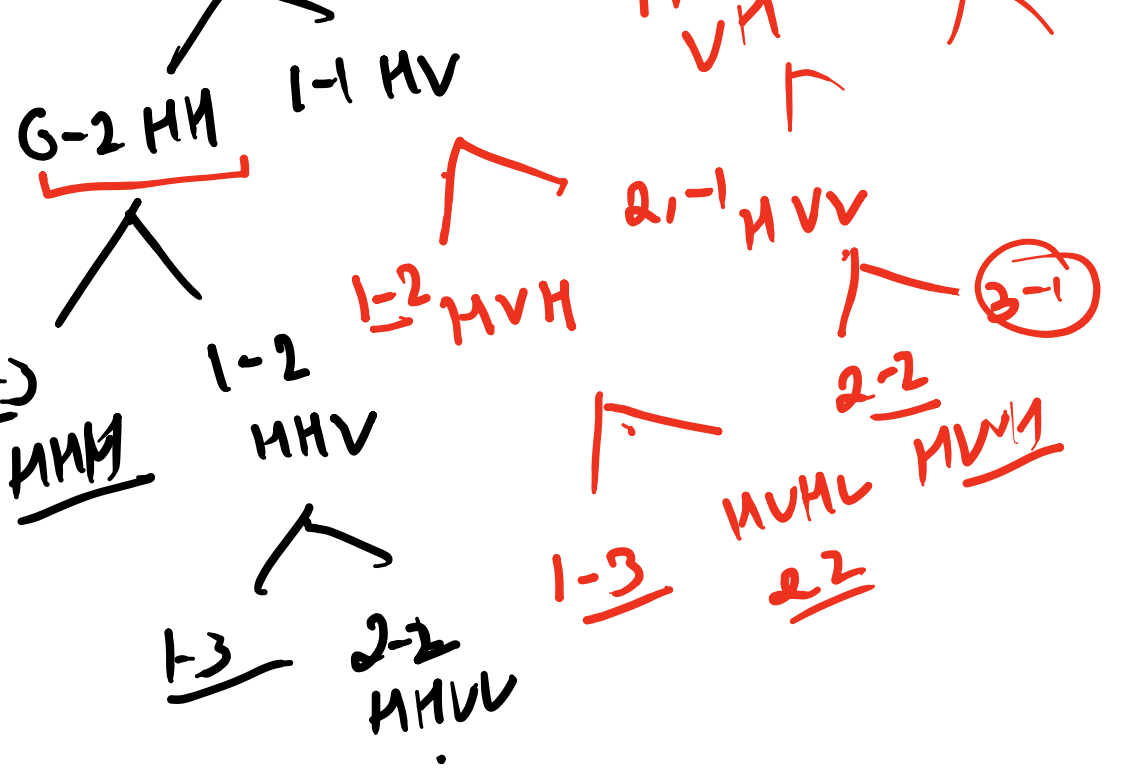
m × n

HW

HVHV

HV VH





[0-0]
0-1 H
1-0 V
0-2 HH
1-1 HV
0-3 HHH
1-2 HHV
1-3
2-2 HHVV
1-2 HVH
2-1 HVV
1-1 VH
2-0 VV
2-2 HVHV
1-3
2-2 HVVH
2-1